



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF COMPUTER NETWORKS AND CYBER SECURITY
CNS 2105: VOICE AND DATA COMMUNICATION ASSIGNMENT 1

Due date: **May 13, 2024**

Time: **1 Hours**

Instructions

1. This examination consists of **TWO** questions. Answer BOTH questions.
2. You may use the formula and Table provided at the end of the paper-page 2

QUESTION ONE

- a. State THREE reasons for modulation in communication networks
- b. Define the following as applied to amplitude modulation
 - i. Baseband signal
 - ii. Passband signal
 - iii. Modulation index
- c. A carrier wave represented by $10 \cos(2\pi 10^6 t) V$ is amplitude modulated by a second wave represented by $3 \cos(2\pi 10^3 t) V$. Calculate
 - i. The depth of modulation
 - ii. The upper and lower side frequencies
 - iii. The amplitude of the side frequencies
 - iv. The fraction of the power transmitted in the sidebands
- d. Draw a block diagram of a pulse code modulation system and state the function of each block

QUESTION TWO

- a. A data sequence 1100001111000010100 is to be encoded using a line code. On the same axis, draw the waveforms for each of the following line encoding schemes:
 - i. Manchester
 - ii. Differential Manchester
 - iii. Bipolar AMI
 - iv. HDB3
- b. Explain the reason for scrambling of data sequence before encoding in data encoding